

03b_centrality

GR

2024-08-22

Load libraries

```
library(igraph)
library(igraphdata)
library(tidyverse)
library(ggnetwork)
```

Create / load several graphs. Save the layouts so they look similar each time we plot them.

```
chain3.g <- make_ring(3) %>% delete_edges("1|3") # create a chain of 3 nodes
chain3.layout <- layout_with_fr(chain3.g)

tree11.g <- make_tree(11, mode="undirected") # create a tree of 11 nodes
tree11.layout <- layout_with_fr(tree11.g)

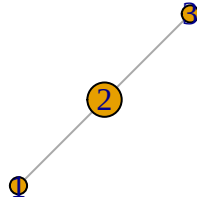
data("kite")
kite.g <- upgrade_graph(kite) %>% igraph::simplify() # load Krackhardt's kite
kite.layout <- layout_with_fr(kite.g)

data("karate")
karate.g <- upgrade_graph(karate) # load Zachary's karate club
karate.layout <- layout_with_fr(karate.g)

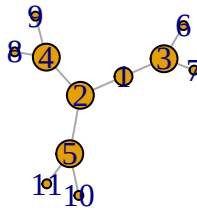
iliad.g <- read_graph("iliad_killed.net", format = "pajek") %>% # load the Iliad combat network
  reverse_edges() # flip the arcs so it is not "A killed B" but "B gave victory to A"
iliad.layout <- layout_with_fr(iliad.g)
```

Degree Centrality

```
chain3.degree <- degree(chain3.g) # create vector of degree centralities for each node in chain3.g
plot_igraph(chain3.g, # name the network object we want to graph
  layout=chain3.layout, # use the saved layout we generated earlier
  vertex.size=chain3.degree*20) # make size proportional to degree centrality. "20" is just t
```

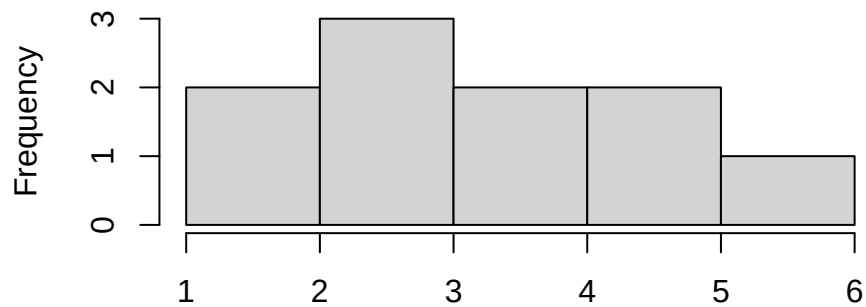


```
tree11.degree <- degree(tree11.g) # create vector of degree centralities for each node in tree11.g
plot.igraph(tree11.g, # name the network object we want to graph
  layout = tree11.layout, # use the saved layout we generated earlier
  vertex.size=tree11.degree*10) # make size proportional to degree centrality. "1.5" is just
```

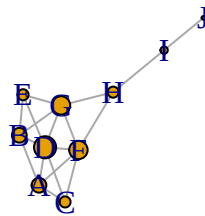


```
kite.degree <- kite.g %>% degree()
kite.degree %>% hist() # note the uniform distribution. does this seem natural?
```

Histogram of .

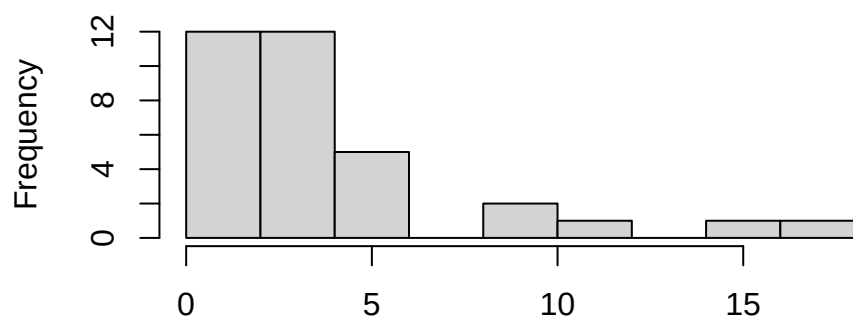


```
plot.igraph(kite.g,
            layout = kite.layout,
            vertex.size=kite.degree*4)
```

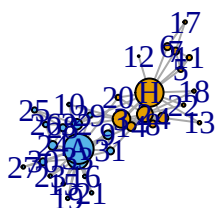


```
karate.degree <- karate.g %>% degree()
karate.degree %>% hist() # note right-skewed dist. what algorithmic network does this resemble?
```

Histogram of .



```
plot.igraph(karate.g,
  layout = karate.layout,
  vertex.size=karate.degree*2)
```



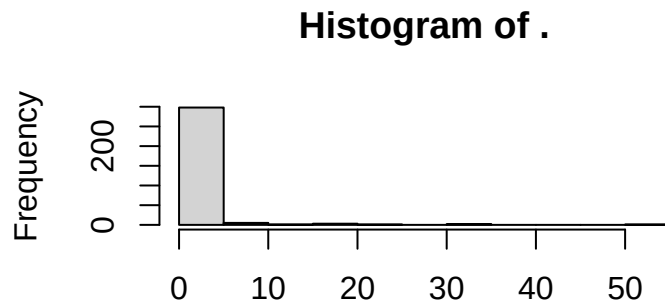
```
iliad.dir.all <- iliad.g %>% degree(mode="all")
iliad.dir.in <- iliad.g %>% degree(mode="in")
iliad.dir.out <- iliad.g %>% degree(mode="out")
iliad.dir.undir <- iliad.g %>% igraph::simplify() %>% degree()

iliad.degree.mat <- cbind(iliad.dir.all, iliad.dir.in, iliad.dir.out, iliad.dir.undir)
iliad.degree.mat %>% cor()
```

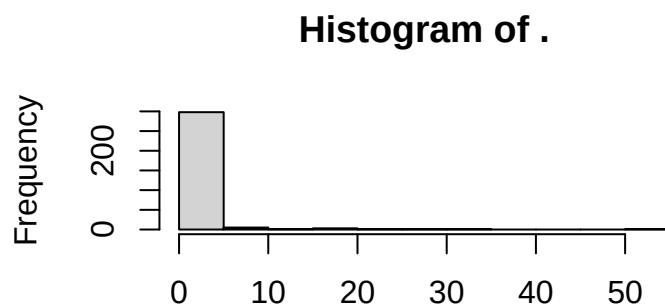
```
##          iliad.dir.all iliad.dir.in iliad.dir.out iliad.dir.undir
## iliad.dir.all      1.00000000  0.9981988  -0.09506425   0.9998553
## iliad.dir.in       0.99819884  1.0000000  -0.15461367   0.9984076
```

```
## iliad.dir.out      -0.09506425   -0.1546137    1.00000000    -0.1009117
## iliad.dir.undir    0.99985528    0.9984076   -0.10091173    1.00000000
```

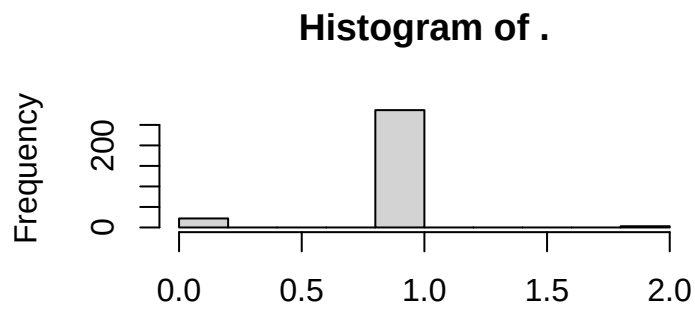
```
iliad.dir.all %>% hist()
```



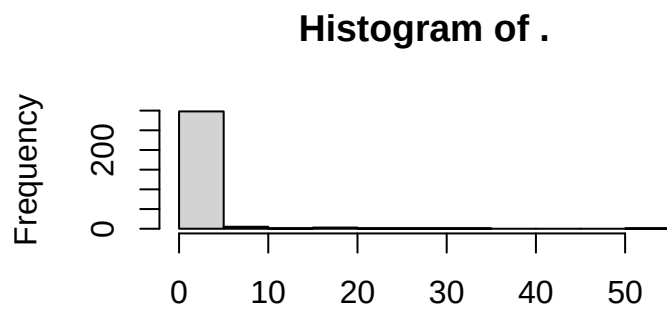
```
iliad.dir.in %>% hist()
```



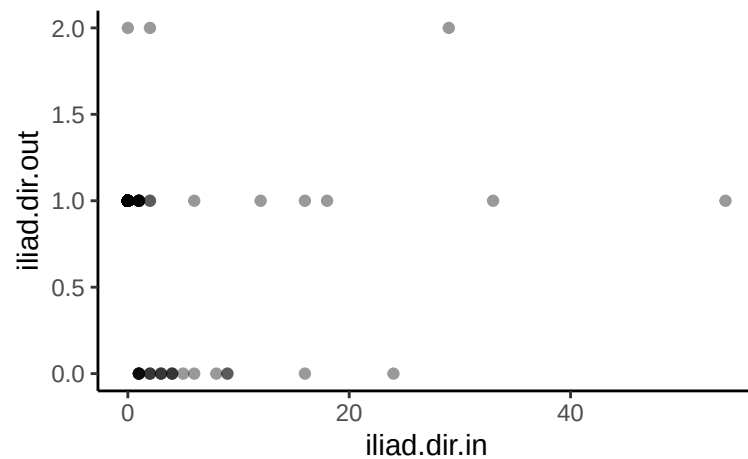
```
iliad.dir.out %>% hist()
```



```
iliad.dir.undir %>% hist()
```



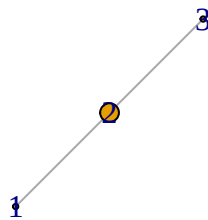
```
iliad.degree.mat %>% as_tibble() %>%
  ggplot(aes(x=iliad.dir.in, y=iliad.dir.out)) +
  geom_point(alpha=0.4) + theme_classic()
```



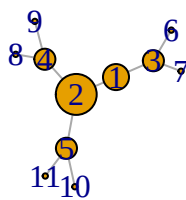
Betweenness Centrality

brokerage

```
chain3.between <- betweenness(chain3.g)
plot.igraph(chain3.g,
  layout = chain3.layout,
  vertex.size=chain3.between*20)
```

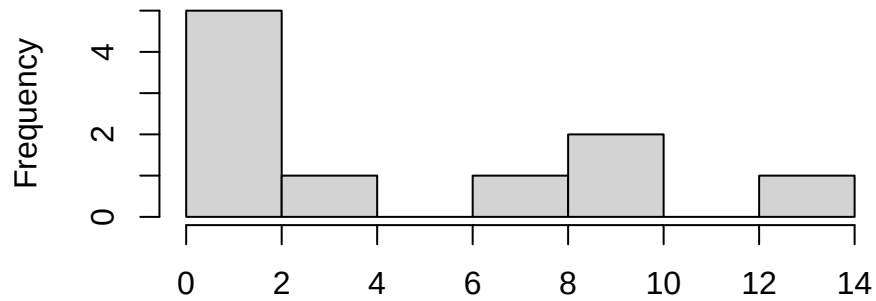


```
tree11.between <- betweenness(tree11.g)
plot.igraph(tree11.g,
  layout = tree11.layout,
  vertex.size=tree11.between*1.5)
```

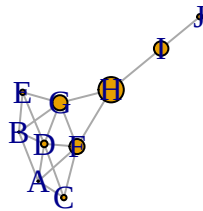


```
kite.betweenness <- kite.g %>% betweenness()
kite.betweenness %>% hist()
```

Histogram of .

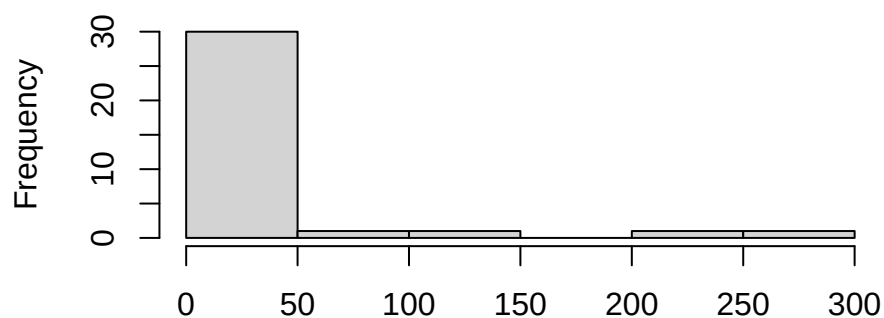


```
plot.igraph(kite.g,
  layout = kite.layout,
  vertex.size=kite.betweenness*2)
```

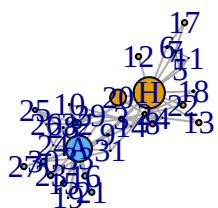


```
karate.betweenness <- karate.g %>% betweenness()
karate.betweenness %>% hist()
```

Histogram of .



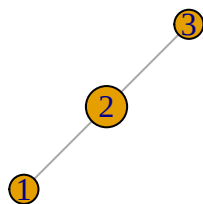
```
plot.igraph(karate.g,
            layout = karate.layout,
            vertex.size=karate.betweenness/7)
```



Eigenvector or Bonacich Centrality

dependence

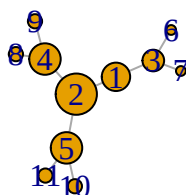
```
chain3.eigen <- eigen_centrality(chain3.g)$vector
plot.igraph(chain3.g,
            layout = chain3.layout,
            vertex.size=chain3.eigen*50)
```



```

tree11.eigen<- eigen centrality(tree11.g)$vector
plot.igraph(tree11.g,
  layout = tree11.layout,
  vertex.size=tree11.eigen*50)

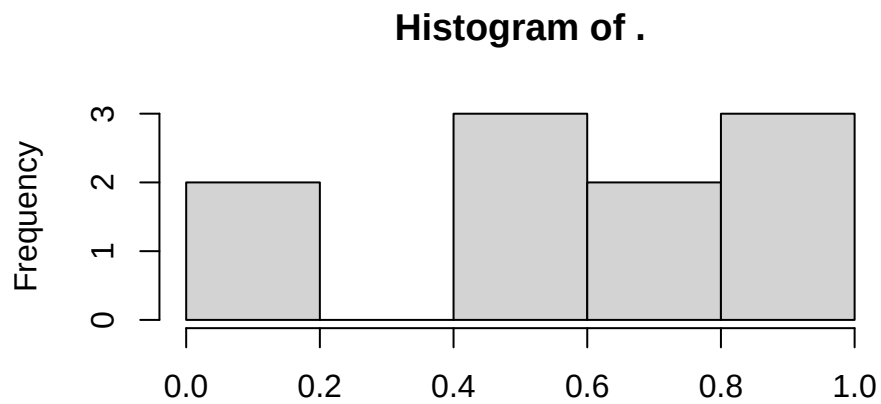
```



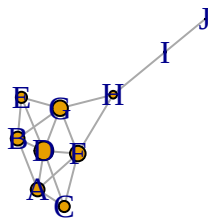
```

kite.eigen <- eigen centrality(kite.g)$vector
kite.eigen %>% hist()

```

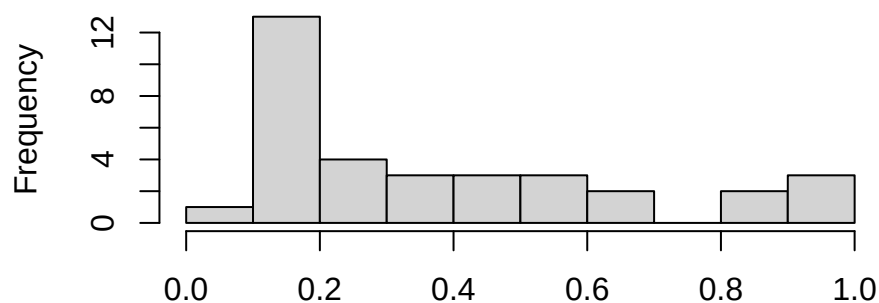


```
#kite.g <- set_vertex_attr(kite.g, 'eigenvector', value=kite.eigen) # how you can save a centrality vec
plot.igraph(kite.g,
  layout = kite.layout,
  vertex.size=kite.eigen*20)
```

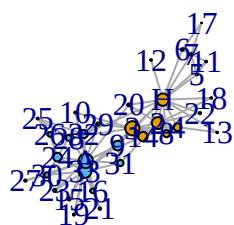


```
karate.eigen <- eigen_centrality(karate.g)$vector
karate.eigen %>% hist()
```

Histogram of .



```
#kite.g <- set_vertex_attr(kite.g, 'eigenvector', value=kite.eigen)
plot.igraph(karate.g,
  layout = karate.layout,
  vertex.size=karate.eigen*15)
```



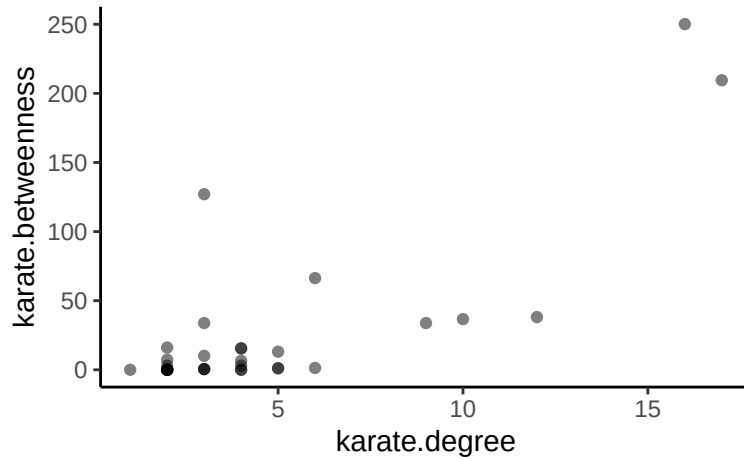
Comparison of Undirected Centralities

```
karate.centralities <- cbind(karate.degree, karate.betweenness, karate.eigen) %>% as_tibble()
karate.centralities %>% cor()
```

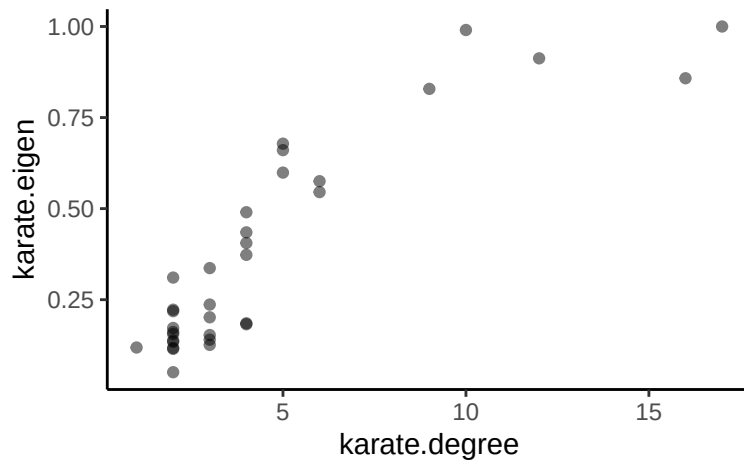
```
##          karate.degree karate.betweenness karate.eigen
## karate.degree      1.000000      0.7882018    0.8783299
```

```
## karate.betweenness    0.7882018        1.0000000    0.5180063
## karate.eigen          0.8783299        0.5180063    1.0000000
```

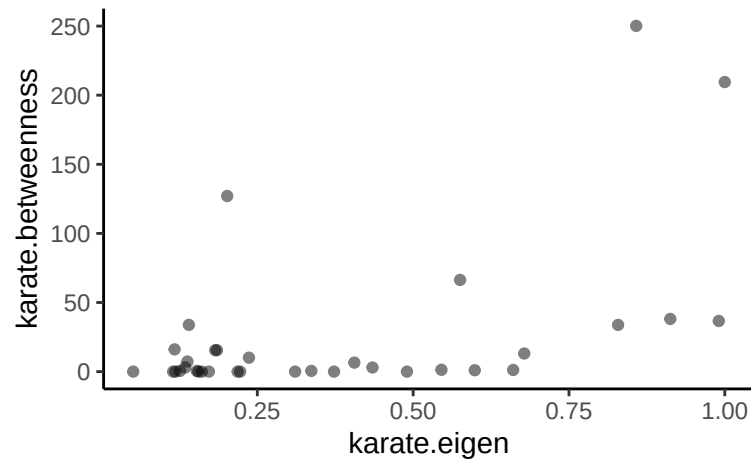
```
karate.centralities %>% ggplot(aes(x=karate.degree,y=karate.betweenness)) +
  geom_point(alpha=0.5) +
  theme_classic()
```



```
karate.centralities %>% ggplot(aes(x=karate.degree,y=karate.eigen)) +
  geom_point(alpha=0.5) +
  theme_classic()
```



```
karate.centralities %>% ggplot(aes(x=karate.eigen,y=karate.betweenness)) +
  geom_point(alpha=0.5) +
  theme_classic()
```

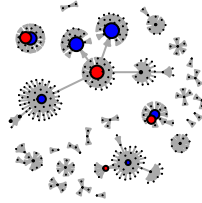


Alpha Centrality and Page Rank

```
iliad.alpha90 <- iliad.g %>% alpha centrality(alpha=.9)
iliad.alpha50 <- iliad.g %>% alpha centrality(alpha=.5)
iliad.pagerank <- iliad.g %>% page_rank()
iliad.indegree <- iliad.g %>% degree(mode = "in")
cbind(iliad.alpha90, iliad.alpha50, iliad.pagerank$vector, iliad.indegree) %>% cor()
```

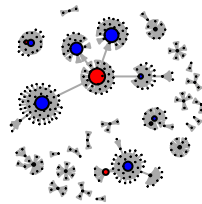
```
##           iliad.alpha90 iliad.alpha50           iliad.indegree
## iliad.alpha90      1.0000000      0.8910417 0.9738803      0.7161362
## iliad.alpha50      0.8910417      1.0000000 0.9133379      0.9054100
##                0.9738803      0.9133379 1.0000000      0.7801648
## iliad.indegree      0.7161362      0.9054100 0.7801648      1.0000000
```

```
plot.igraph(iliad.g,
  vertex.label=NA,
  layout = iliad.layout,
  vertex.size = iliad.alpha90/7,
  edge.arrow.size=0.2)
```

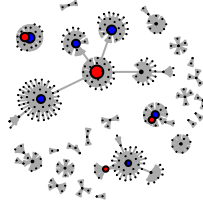


```
# Ajax, Hector, Achilles
```

```
plot.igraph(iliad.g,
  vertex.label=NA,
  layout = iliad.layout,
  vertex.size = iliad.alpha50/2,
  edge.arrow.size=0.2)
```



```
plot.igraph(iliad.g,
  vertex.label=NA,
  layout = iliad.layout,
  vertex.size = iliad.pagerank$vector*150,
  edge.arrow.size=0.2)
```



```
plot.igraph(iliad.g,
  vertex.label=NA,
  layout = iliad.layout,
  vertex.size = iliad.indegree/2,
  edge.arrow.size=0.2)
```

